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Multilingual Home Automation System Using Google Assistant And Raspberry Pi

One of the problems with a voice-controlled smart home automation system is the language barrier. Most of the existing home automation systems support only a few languages, including English.

In this DIY project, we are going

to make an Internet of Things (IoT) home automation system that allows you to control your home devices with voice commands in your preferred language like Hindi, Bengali, Marathi, and so on.

It uses Google Assistant to control smart devices at home and also sup-

After successful installation of the server, check it by typing the IP address of your Raspberry Pi in any web browser. If the server is working fine, you will get the Apache page in the web browser.

Now, use your Apache server to process PHP files. For this, you need to have the latest version of the PHP module for Apache. To get this version, run this command:

```
sudo apt-get install php libapache2-mod-php -y
```

Next, create a PHP file to control the GPIO of Raspberry Pi. To do so, open the terminal in Raspberry Pi and go to its HTML directory using the

```
pi@raspberrypi: /var/www/html
File Edit Tabs Help
gdb mailcap.order rcS.d xdg
ghostscript manpath.config request-key.conf xml
glvnd matplotlibrc request-key.d
gnome menu-methods resolv.conf
groff mime.types resolvconf
pi@raspberrypi:/etc $ cd html
bash: cd: html: No such file or directory
pi@raspberrypi:/etc $ cd ../
pi@raspberrypi:/$ $ ls
bin dev home lost+found mnt proc run srv tmp var
boot etc lib media opt root sbin sys usr
pi@raspberrypi:/ $ cd var/
pi@raspberrypi:/var $ ls
backups cache lib local lock log mail opt run spool swap tmp www
pi@raspberrypi:/var $ cd www/
pi@raspberrypi:/var/www $ ls
html
pi@raspberrypi:/var/www $ cd html/
pi@raspberrypi:/var/www/html $ ls
coupon.php efy.jpg lighthoff.php page.php submit3.php test2.php
data.txt form.php page1.php Penguins.jpg submit4.php upload.php
dcd.jpg index.html page2.php story.php submit.php
dwd.jpg lightson.php page2.php submit2.php test1.php
pi@raspberrypi:/var/www/html $ sudo nano lightson.php
```

Fig. 1: Creating PHP file from terminal

```
<?php
System ("gpio -g mode 27 out");
System ("gpio -g write 27 1");
?>
```

Fig. 2: PHP code

Components Table 1		
Component	Quantity	Description
Raspberry Pi Zero W	1	For programming
5V relay module (2-channel)	1	For controlling appliances
Jumpers wires	4	Female to female
Hook-up wires	20cm	For connection
5V power adaptor	1	AC to DC adaptor

ports all the languages supported by Google Assistant. We will also make an Android app for a smart touch control system.

Let's first start the voice command control project. The components required for the project are listed in Table 1.

The first thing is to set up an Apache server in Raspberry Pi. For that, open the terminal window and run the following command:

```
sudo apt-get install apache2 -y
```

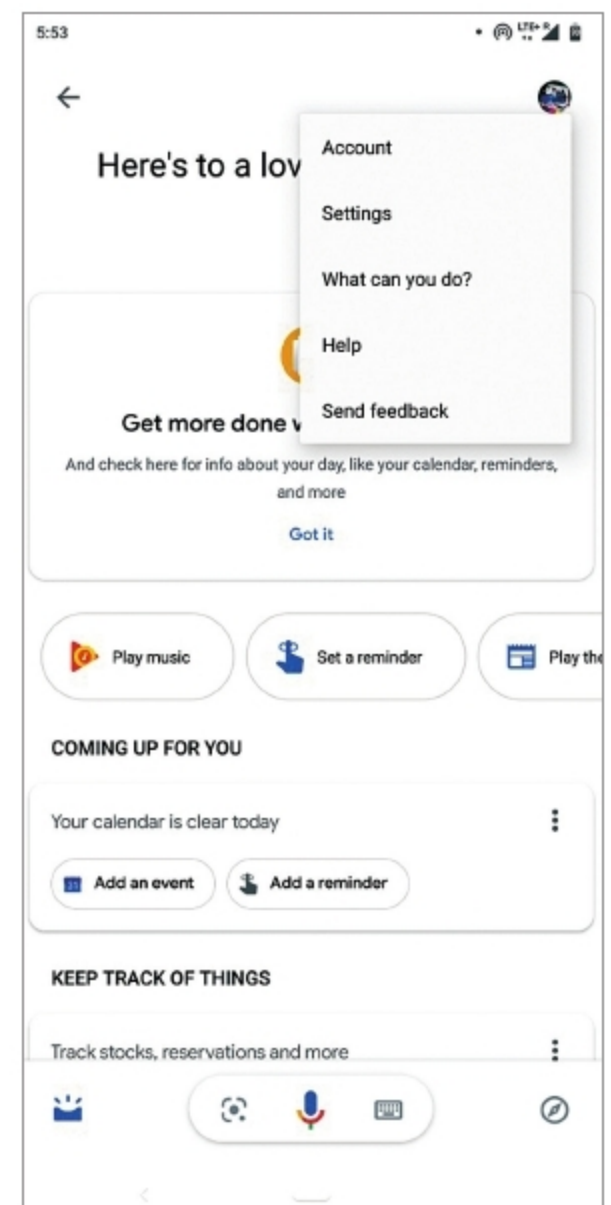


Fig. 3: Settings in Google Assistant